

MATH 350 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 1
 - Foundations
 - The tenth axiom

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Defining the reals

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let's define these things

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Notation: $z := x/y$

Special notation: $0 := x - x$ and $1 := x/x$, and $-x := 0 - x$

Challenge!

Problem

use the axioms to show that $(x + y)z = xz + yz$.

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use the axioms to show that $(x + y)z = xz + yz$.

Solution

$$(x + y)z \stackrel{A1}{=} z(x + y) \stackrel{A3}{=} zx + zy \stackrel{A1}{=} xz + yz.$$

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use the axioms to show that the value of $x - x$ doesn't depend on x .

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Suppose $z_1 = x - x$ and $z_2 = y - y$.
Then $x = x + z_1$ and $y = y + z_2$.

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Solution

Suppose $z_1 = x - x$ and $z_2 = y - y$.

Then $x = x + z_1$ and $y = y + z_2$. Therefore

$$(x + y) + z_1 \stackrel{A2}{=} x + (y + z_1) \stackrel{A1}{=} x + (z_1 + y) \stackrel{A2}{=} (x + z_1) + y = x + y,$$

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Suppose $z_1 = x - x$ and $z_2 = y - y$.

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$$(x + y) + z_2 \stackrel{A2}{=} x + (y + z_2) = x + y,$$

so that

$$z_1 = (x + y) - (x + y) = z_2.$$

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Suppose $z_1 = x/x$ and $z_2 = y/y$.
Then $x = z_1 x$ and $y = z_2 y$.

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Solution

Suppose $z_1 = x/x$ and $z_2 = y/y$.
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so that

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- **field of Boolean numbers**

$$\mathbb{F}_2 = \{0, 1\}$$

$$0 + 0 = 0, \quad 0 + 1 = 1, \quad 1 + 0 = 1, \quad 1 + 1 = 0$$

$$0 \cdot 0 = 0, \quad 0 \cdot 1 = 0, \quad 1 \cdot 0 = 0, \quad 1 \cdot 1 = 1$$

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Solution

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Solution

There aren't multiplicative inverses! For example, $1/2$ doesn't make sense because there isn't an integer z with $1 = 2z$.

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Special notation:

- $x > y$ means $y < x$
- $x \leq y$ means $x < y$ or $x = y$
- $x \geq y$ means $y \leq x$

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but this means both $0 < 1$ and $1 < 0$, which violates the trichotomy.

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but this means both $0 < 1$ and $1 < 0$, which violates the trichotomy. Same outcome if $1 < 0$.

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This says $a < a$, which contradicts the trichotomy. □

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Real numbers

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- Intuitively, it represents the fact that the real line has no holes or gaps.

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- 5 is a maximal element of $[-1, 5]$
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- $\mathbb{Z}_+$ has no upper bound
- $[3, 7)$ has an upper bound but no maximal element

Challenge!

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Show that if S has a maximal element, then it is unique.

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Hint: use the definition!

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Suppose that b_1 and b_2 are both maximal elements of S .

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Suppose that b_1 and b_2 are both maximal elements of S .
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.

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Solution

Suppose that b_1 and b_2 are both maximal elements of S .

Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.

Moreover, $b_1 \in S$ and $b_2 \in S$.

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Moreover, $b_1 \in S$ and $b_2 \in S$.

Since $b_1 \in S$ and b_2 is an upper bound of S , $b_1 \leq b_2$.

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Likewise, since $b_2 \in S$ and b_1 is an upper bound of S , $b_2 \leq b_1$.

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By the trichotomy, we find $b_1 = b_2$.

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By the trichotomy, we find $b_1 = b_2$.

Now we can say *the* maximum, $\max(S)$

Supremum

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In other words

a supremum is a **least upper bound**

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Solution

Suppose that b_1 and b_2 are both suprema of S .
Then b_1 and b_2 are both upper bounds of S .
Since b_1 is a least upper bound, $b_1 \leq b_2$.

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Then b_1 and b_2 are both upper bounds of S .

Since b_1 is a least upper bound, $b_1 \leq b_2$.

Since b_2 is also a least upper bound, $b_2 \leq b_1$.

By the trichotomy, we find $b_1 = b_2$.

Now we can say *the* supremum, $\sup(S)$

Challenge!

Problem

Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

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Let $b = \max(S)$.

Then b is an upper bound of S and $b \in S$.

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If b' is another upper bound of S , then by definition $b \leq b'$.

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If b' is another upper bound of S , then by definition $b \leq b'$.

Thus b is the least upper bound of S .

Completeness Axiom

Now we have the machinery in place to state the Completeness Axiom.

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Examples:

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- 1 is the supremum of

$$\{\sin(n) : n \in \mathbb{Z}_+\}$$